

Subsystem A

$$\dot{x}_A = f_A(x_A, u_A, t)$$

$$y_A = h_A(x_A, u_A)$$

$$u_B = y_A$$

Subsystem B

$$\dot{x}_B = f_B(x_B, u_B, t)$$

$$y_B = h_B(x_B, u_B)$$

$$u_A = y_B$$